EXPERIMENT 03

Morphological Analysis using NLTK Library

AIM

To perform morphological analysis on words and sentences using the Natural Language Toolkit (NLTK) in Python, analyzing affixes, base stems, and lemmas.

PROCEDURE

1. Import `nltk` and required modules (WordNetLemmatizer, pos_tag, word_tokenize).
2. Tokenize input text into constituent lexical items.
3. Extract root morphemes using lemmatization coupled with Part-of-Speech context.
4. Examine grammatical morphemes (prefixes, suffixes, inflections) by comparing surface forms to base forms.
5. Display the morphological breakdown and base lemmas.

PROGRAM

```
import nltk
from nltk.stem import WordNetLemmatizer
from nltk.corpus import wordnet

nltk.download('wordnet', quiet=True)
nltk.download('averaged_perceptron_tagger', quiet=True)
nltk.download('punkt', quiet=True)

lemmatizer = WordNetLemmatizer()

def get_wordnet_pos(treebank_tag):
    if treebank_tag.startswith('J'):
        return wordnet.ADJ
    elif treebank_tag.startswith('V'):
        return wordnet.VERB
    elif treebank_tag.startswith('N'):
        return wordnet.NOUN
    elif treebank_tag.startswith('R'):
        return wordnet.ADV
    else:
        return wordnet.NOUN

words = ["cats", "running", "better", "corpora", "studies", "flying", "easily"]
print(f"{'Word':<12} | {'POS Tag':<8} | {'Base Lemma':<12} | {'Morphological
Feature'}")
print("-" * 55)

tagged = nltk.pos_tag(words)
for word, tag in tagged:
    wn_pos = get_wordnet_pos(tag)
```

```
    lemma = lemmatizer.lemmatize(word, pos=wn_pos)
    feature = "Plural Noun" if word.endswith('s') and wn_pos == wordnet.NOUN else \
              "Inflected Verb (Progressive)" if word.endswith('ing') else \
              "Comparative Adjective" if word == 'better' else "Base/Inflected Form"
    print(f"{word:<12} | {tag:<8} | {lemma:<12} | {feature}")
```

OUTPUT

```
> python3 experiment_03.py
-----------------------------------------------------------------------------------------------------
Word         | POS Tag  | Base Lemma   | Morphological Feature
-----------------------------------------------------
cats         | NNS      | cat          | Plural Noun
running      | VBG      | run          | Inflected Verb (Progressive)
better       | JJR      | good         | Comparative Adjective
corpora      | NN       | corpus       | Base/Inflected Form
studies      | NNS      | study        | Plural Noun
flying       | VBG      | fly          | Inflected Verb (Progressive)
easily       | RB       | easily       | Base/Inflected Form
[Process completed with exit code 0]
```

RESULT

Thus, morphological analysis was successfully conducted using the NLTK library, accurately deriving base lemmas and grammatical morphemes.